

Yinhua Piao

PERSONAL INFORMATION

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Homepage: <http://qkrmsgkhk.github.io/>
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EDUCATION & EXPERIENCE

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| 2025.8–Present | Postdoc Fellow at Innocore AI-CRED Institute
KAIST, Seoul, Korea.
Advisor: Prof. Sungsoo Ahn, Prof. Woo Youn Kim
Research: Computational Cell Modeling with LLM Agents. |
| 2025–2025.7 | Postdoc at Bioinformatics Institute
Seoul National University, Seoul, Korea.
Advisor: Prof. Sun Kim |
| 2020–2025 | Ph.D in Computer Science and Engineering
Seoul National University, Seoul, Korea.
Advisor: Prof. Sun Kim
Thesis: Subgraph-informed Hierarchical Learning in Clinical and Biomedical Domains. (<i>Outstanding Doctoral Dissertation Award</i>) |
| 2018–2020 | M.S. in Computer Science and Engineering
Seoul National University, Seoul, Korea.
Advisor: Prof. Sun Kim
Thesis: Graph Convolutional Networks for Predictive Healthcare using Clinical Notes. |
| 2014–2018 | B.S. in Software Engineering
Jilin University, Changchun, China. |

RESEARCH OVERVIEW

I develop graph-based and multi-modal AI methods for drug discovery and biomedical reasoning, integrating molecular, cellular, sequential, textual, and knowledge-graph signals across the preclinical pipeline, with a growing focus on LLM-based scientific reasoning over biological knowledge.

KAIST (Postdoc) **Computational Cell Modeling**

- **Genetic perturbation prediction**[1] — Gene expression $\times$ Gene descriptions $\times$ Biological KG
- **Perturbation reasoning agent**[2] — LLM agent $\times$ Progressive reasoning

SNU (Ph.D) **Graph AI in Preclinical Drug Discovery** (Integrating molecular sequence and structure, gene expression, clinical text, and biological knowledge graphs)

- **Chemical reaction prediction**[3] — Molecular structure $\times$ Context
- **Molecule understanding**[4] — Molecular sequence/structure $\times$ Text
- **Drug-likeness prediction**[5] — Biological KG $\times$ Molecular structure
- **OOD molecular property prediction**[6] — Molecular structure $\times$ Invariant learning
- **Drug response prediction**[9] — Molecular structure $\times$ Gene expression $\times$ Biological KG
- **Patient Stratification**[7, 8] — EHR clinical notes $\times$ Hierarchical graph learning

HONORS, AWARDS & SCHOLARSHIPS

Aug. 2025	Innocore AI-Cred Postdoctoral Fellowship, KAIST.
Feb. 2025	Outstanding Doctoral Dissertation Award, SNU.
Dec. 2024	Research Fair Best Poster Award, SNU.
Aug. 2024	Youlchon AI Star Scholarship.
Feb. 2024	Samsung HumanTech Paper Award.
Apr. 2022	AIIS Retreat Best Poster Award.
Feb. 2022	Samsung HumanTech Paper Award (Bronze Prize).
Feb. 2022	AAAI Student Scholar Scholarship.
Jun.20–Jan.22	SNU Global Scholarships II.

SELECTED PUBLICATIONS

1. PIAO, Y., KIM, H., KIM, S., OH, Y., JEON, J., HWANG, S.-Y., LIM, J., KIM, W. Y., PARK, C., AND AHN, S. Learning adaptive perturbation-conditioned contexts for robust transcriptional response prediction. In *Forty-third International Conference on Machine Learning* (2026)

2. KIM, H., HWANG, S.-Y., LIM, J., PIAO, Y., OH, Y., KIM, W. Y., PARK, C., AHN, S., AND JEON, J. Progressive multi-agent reasoning for biological perturbation prediction. *arXiv preprint arXiv:2602.07408* (2026)
3. LU, Y., PIAO, Y., SHEN, M., AND KIM, S. Improving reaction yield prediction with chemical atom-level reaction learning. *Journal of Chemical Information and Modeling* 65, 21 (2025), 11572–11582
4. HA, S., KIM, J. H., PIAO, Y., CHO, C., AND KIM, S. MV-CLAM: Multi-view molecular interpretation with cross-modal projection via language model. In *Findings of the Association for Computational Linguistics: EMNLP 2025* (Suzhou, China, Nov. 2025), C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds., Association for Computational Linguistics, pp. 21528–21549
5. BANG, D., SUNG, I., PIAO, Y., LEE, S., AND KIM, S. Predicting drug-likeness via biomedical knowledge alignment and EM-like one-class boundary optimization. In *Forty-second International Conference on Machine Learning* (2025)
6. PIAO, Y., LEE, S., LU, Y., AND KIM, S. Improving out-of-distribution generalization in graphs via hierarchical semantic environments. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2024)
7. *KIM, N., *PIAO, Y., AND KIM, S. Clinical note owns its hierarchy: Multi-level hypergraph neural networks for patient-level representation learning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Toronto, Canada, July 2023), A. Rogers, J. Boyd-Graber, and N. Okazaki, Eds., Association for Computational Linguistics, pp. 5559–5573
8. PIAO, Y., LEE, S., LEE, D., AND KIM, S. Sparse structure learning via graph neural networks for inductive document classification. In *Proceedings of the AAAI conference on artificial intelligence* (2022), vol. 36, pp. 11165–11173
9. *SHIN, J., *PIAO, Y., BANG, D., KIM, S., AND JO, K. Drpreter: Interpretable anticancer drug response prediction using knowledge-guided graph neural networks and transformer. *International Journal of Molecular Sciences* 23, 22 (2022), 13919

PUBLICATIONS

1. PIAO, Y., KIM, H., KIM, S., OH, Y., JEON, J., HWANG, S.-Y., LIM, J., KIM, W. Y., PARK, C., AND AHN, S. Learning adaptive perturbation-conditioned contexts for robust transcriptional response prediction. In *Forty-third International Conference on Machine Learning* (2026)
2. LU, Y., PIAO, Y., SHEN, M., AND KIM, S. Improving reaction yield prediction with chemical atom-level reaction learning. *Journal of Chemical Information and Modeling* 65, 21 (2025), 11572–11582
3. HA, S., KIM, J. H., PIAO, Y., CHO, C., AND KIM, S. MV-CLAM: Multi-view molecular interpretation with cross-modal projection via language model. In *Findings of the Association for*

- Computational Linguistics: EMNLP 2025* (Suzhou, China, Nov. 2025), C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds., Association for Computational Linguistics, pp. 21528–21549
4. KIM, Y., PIAO, Y., LEE, S., AND KIM, S. Aligning molecules and fragments in a shared embedding space for RL-based molecule generation. In *ICLR 2025 Workshop on Machine Learning for Genomics Explorations* (2025)
 5. BANG, D., SUNG, I., PIAO, Y., LEE, S., AND KIM, S. Predicting drug-likeness via biomedical knowledge alignment and EM-like one-class boundary optimization. In *Forty-second International Conference on Machine Learning* (2025)
 6. LU, Y., PIAO, Y., LEE, S., AND KIM, S. Context-aware hierarchical fusion for drug relational learning. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* (2025)
 7. CHO, C., LEE, S., BANG, D., PIAO, Y., AND KIM, S. Chemap: predicting drug approval with chemical structures before clinical trial phase by leveraging multi-modal embedding space and knowledge distillation. *Scientific Reports* 14, 1 (2024), 23010
 8. LEE, S., PARK, J., PIAO, Y., LEE, D., LEE, D., AND KIM, S. Multi-layered knowledge graph neural network reveals pathway-level agreement of three breast cancer multi-gene assays. *Computational and Structural Biotechnology Journal* (2024)
 9. LU, Y., PIAO, Y., AND KIM, S. Enhancing drug-drug interaction prediction with context-aware architecture. In *The Second Tiny Papers Track at ICLR 2024*
 10. PIAO, Y., LEE, S., LU, Y., AND KIM, S. Improving out-of-distribution generalization in graphs via hierarchical semantic environments. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2024)
 11. *KIM, N., *PIAO, Y., AND KIM, S. Clinical note owns its hierarchy: Multi-level hypergraph neural networks for patient-level representation learning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Toronto, Canada, July 2023), A. Rogers, J. Boyd-Graber, and N. Okazaki, Eds., Association for Computational Linguistics, pp. 5559–5573
 12. YI, J., LEE, S., LIM, S., CHO, C., PIAO, Y., YEO, M., KIM, D., KIM, S., AND LEE, S. Exploring chemical space for lead identification by propagating on chemical similarity network. *Computational and Structural Biotechnology Journal* 21 (2023), 4187–4195
 13. PIAO, Y., LEE, S., LEE, D., AND KIM, S. Sparse structure learning via graph neural networks for inductive document classification. In *Proceedings of the AAAI conference on artificial intelligence* (2022), vol. 36, pp. 11165–11173
 14. *SHIN, J., *PIAO, Y., BANG, D., KIM, S., AND JO, K. Drpreter: Interpretable anticancer drug response prediction using knowledge-guided graph neural networks and transformer. *International Journal of Molecular Sciences* 23, 22 (2022), 13919
 15. LIM, S., LEE, S., PIAO, Y., CHOI, M., BANG, D., GU, J., AND KIM, S. On modeling and utilizing chemical compound information with deep learning technologies: A task-oriented approach. *Computational and Structural Biotechnology Journal* 20 (2022), 4288–4304
 16. LEE, S., LEE, D., PIAO, Y., AND KIM, S. Spgp: Structure prototype guided graph pooling. In *NeurIPS 2022 Workshop: New Frontiers in Graph Learning* (2022)

17. BAE, J. H., SHIN, Y., PIAO, Y., LIM, J. H., AND KIM, S. Mo1641: Confidence estimation in clinical decision support system for determination of colonoscopy surveillance interval. *Gastroenterology* 162, 7 (2022), S-839
18. KIM, S., BAE, S., PIAO, Y., AND JO, K. Graph convolutional network for drug response prediction using gene expression data. *Mathematics* 9, 7 (2021), 772
19. PIAO, Y. Graph convolutional networks for predictive healthcare using clinical notes, 2020

COMMUNITY SERVICE

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|-----------|---|
| 2022-2026 | (Proceedings) AAAI, IJCAI, NeurIPS, CVPR, KDD, ACML, ICCV, ACL, ICML. |
| 2022-2026 | (Journal) BMC Bioinformatics, BMC Biology, BMC Medical Informatics and Decision Making, Npj Systems Biology and Applications, Pattern Analysis and Applications, Scientific Reports, Transactions on Knowledge Discovery from Data, Journal of Biomedical Informatics, Computational and Structural Biotechnology Journal, Journal of Cheminformatics, BioData Mining, Discover Molecules, Knowledge and Information Systems. |

PRESENTATIONS

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| April. 2026 | Invited Talk , Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction.
<i>The Multiomics Reading Group, Valence Lab</i> |
| Sep. 2025 | Invited Talk , Improving out-of-distribution generalization in graphs via hierarchical semantic environments.
<i>Dongkuk University</i> |
| Jan. 2025 | Invited Talk , Improving out-of-distribution generalization in graphs via hierarchical semantic environments.
<i>AI/Computing Frontier School in Seoul National University</i> |
| Dec. 2023 | Invited Talk , Multi-level Hypergraph Neural Networks for Patient-level Representation Learning.
<i>ModuleCon, ModuleLab in Ewha Womans University</i> |
| Dec. 2023 | Invited Talk , Sparse Structure learning via GNNs for Document Classification.
<i>ModuleCon, ModuleLab in Ewha Womans University</i> |
| Nov. 2022&2023 | Tutorial Self-supervised Learning on Molecule Graphs.
<i>AI-BIO Researcher Training Course, AIIS in Seoul National University</i> |
| Sep. 2022 | Invited Talk , Sparse Structure learning via GNNs for Document Classification.
<i>CSE Fun Lunch Seminars in Seoul National University</i> |
| Apr. 2022 | Invited Talk , Sparse Structure learning via GNNs for Document Classification
<i>AI Retreat, AIIS in Seoul National University</i> |

TEACHING

Spring 2025	Teaching Assistant , Machine Learning on Bioinformatics.
Fall 2024	Head Teaching Assistant , Algorithms.
Spring 2024	Head Teaching Assistant , Machine Learning on Bioinformatics.
Fall 2023	Teaching Assistant , Algorithms.
Spring 2023	Teaching Assistant , Machine Learning on Bioinformatics.
Fall 2022	Teaching Assistant , Algorithms.
Spring 2022	Teaching Assistant , Machine Learning on Bioinformatics..
Fall 2020	Teaching Assistant , Algorithms.

SUPERVISION

2024-2025	Junha Kim Master Student, SNU. On the process of molecule generation using subgraph mining.
2023-2024	Sumin Ha & Jun Hyeong Kim , Master Student, SNU. The resulting paper is presented at EMNLP 2025.
2023	Sangyeup Kim , Master Student, SNU. The resulting paper is on arXiv.
2022-2023	Nayeon Kim , Master Student, SNU. The resulting paper got an oral presentation (top 0.7%) at ACL 2023.
2021-2022	Jihye Shin , Master Student, SNU. The resulting paper got accepted by IJMS 2022.

LANGUAGES

Chinese and Korean: Native
English: Fluent

June 15, 2026